The Pandrosion Discriminant: $D(P)^2 = \prod E_P(\alpha_k)$ and the Diagonal Interaction Matrix

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Abstract

We show that the discriminant of a monic polynomial P with simple roots $\alpha_1, \dots, \alpha_d$ satisfies $D(P)^2 = \prod_{k=1}^{d} E_P(\alpha_k)$, where $E_P(\alpha_k) = P'(\alpha_k)^2$ is the Pandrosion energy at the root α_k (Paper 20). The key structural result is that the *Pandrosion interaction matrix* $\mathbf{Q}_{ij} = Q(\alpha_i, \alpha_j)$ is **diagonal**: $Q(\alpha_i, \alpha_j) = \delta_{ij} \cdot P'(\alpha_i)$, with determinant $\det \mathbf{Q} = (-1)^{d(d-1)/2} D(P)$. This places the discriminant as the fundamental invariant of the Pandrosion spectrum.

1 The Pandrosion interaction matrix

Definition 1.1 (Interaction matrix)**.** For $P(z) = \prod_{k=1}^{d}(z - \alpha_k)$ with simple roots, the **Pandrosion interaction matrix** is the $d \times d$ matrix:

$$\mathbf{Q}_{ij} := Q(\alpha_i, \alpha_j) = \prod_{\substack{m=1 \\ m \neq i}}^{d} (\alpha_j - \alpha_m). \tag{1}$$

Theorem 1.2 (Diagonality)**.** *The interaction matrix is diagonal:*

$$Q(\alpha_i, \alpha_j) = \delta_{ij} \cdot P'(\alpha_i). \tag{2}$$

Proof. For $i \neq j$: the product $Q(\alpha_i, \alpha_j) = \prod_{m \neq i}(\alpha_j - \alpha_m)$ ranges over all $m \neq i$. Since $j \neq i$, $m = j$ is included, contributing the factor $\alpha_j - \alpha_j = 0$. Hence $Q(\alpha_i, \alpha_j) = 0$.

For $i = j$: $Q(\alpha_i, \alpha_i) = \prod_{m \neq i}(\alpha_i - \alpha_m) = P'(\alpha_i)$. □

Remark 1.3. The diagonality of $\mathbf{Q}$ is a purely combinatorial consequence of the product structure: any off-diagonal entry contains a "self-cancellation" factor $\alpha_j - \alpha_j = 0$. This is the root-level manifestation of the fact that distinct roots do not interact through the Pandrosion quotient.

2 The discriminant identities

Theorem 2.1 (Pandrosion discriminant)**.** *For a monic polynomial P of degree d with simple roots:*

$$\boxed{D(P) = (-1)^{d(d-1)/2} \prod_{k=1}^{d} Q(\alpha_k, \alpha_k) = (-1)^{d(d-1)/2} \det \mathbf{Q}.} \tag{3}$$

Proof. The discriminant is $D(P) = \prod_{i<j}(\alpha_i - \alpha_j)^2$. The product of all $P'(\alpha_k)$ is:

$$\prod_{k=1}^{d} P'(\alpha_k) = \prod_{k=1}^{d} \prod_{m \neq k} (\alpha_k - \alpha_m) = \prod_{k \neq m} (\alpha_k - \alpha_m).$$

Each unordered pair $\{i, j\}$ contributes $(\alpha_i - \alpha_j)(\alpha_j - \alpha_i) = -(\alpha_i - \alpha_j)^2$, so:

$$\prod_{k \neq m} (\alpha_k - \alpha_m) = \prod_{i<j} \left[-(\alpha_i - \alpha_j)^2\right] = (-1)^{\binom{d}{2}} D(P).$$

Therefore $D(P) = (-1)^{d(d-1)/2} \prod_k P'(\alpha_k) = (-1)^{d(d-1)/2} \det \mathbf{Q}$. ☐

Corollary 2.2 (Energy product formula)**.**

$$\boxed{D(P)^2 = \prod_{k=1}^{d} E_P(\alpha_k) = \prod_{k=1}^{d} P'(\alpha_k)^2.} \tag{4}$$

Proof. $D(P)^2 = \left(\prod_k P'(\alpha_k)\right)^2 = \prod_k P'(\alpha_k)^2 = \prod_k E_P(\alpha_k)$. ☐

3 The vanishing criterion

Proposition 3.1 (Discriminant vanishing)**.** *The following are equivalent:*

(i) $D(P) = 0$ *(repeated root);*

(ii) $\det \mathbf{Q} = 0$ *(interaction matrix is singular);*

(iii) $\exists k : Q(\alpha_k, \alpha_k) = 0$ *(zero diagonal entry);*

(iv) $\exists k : E_P(\alpha_k) = 0$ *(zero Pandrosion energy at a root).*

Proof. Since $\mathbf{Q}$ is diagonal, $\det \mathbf{Q} = \prod_k Q(\alpha_k, \alpha_k) = 0$ iff some $Q(\alpha_k, \alpha_k) = P'(\alpha_k) = 0$ iff α_k is a repeated root iff $E_P(\alpha_k) = P'(\alpha_k)^2 = 0$. ☐

Remark 3.2 (Connection to Casas–Alvero, Paper 12)*.* The Casas–Alvero conjecture states: if P shares a root with $P^{(k)}$ for $k = 1, \ldots, d-1$, then $P = (z - \alpha)^d$. In Pandrosion language (Paper 12): the Q-chain degenerates at every level. The discriminant criterion (iv) is the *first level* ($k = 1$) of this chain: $E_P(\alpha_k) = 0$ detects a shared root between P and P'. Casas–Alvero requires all $d - 1$ levels to degenerate simultaneously.

4 The resultant as a Pandrosion product

Proposition 4.1 (Resultant–discriminant relation)**.** *For monic P with roots $\alpha_1, \ldots, \alpha_d$:*

$$\operatorname{Res}(P, P') = \prod_{k=1}^{d} P'(\alpha_k) = \prod_{k=1}^{d} Q(\alpha_k, \alpha_k) = \det \mathbf{Q}. \tag{5}$$

Proof. The resultant of P and P' equals $\prod_k P'(\alpha_k)$ (standard formula), which equals $\det \mathbf{Q}$ by Theorem 1.2. ☐

For two arbitrary polynomials P (roots α_i) and R (roots β_j), the resultant $\operatorname{Res}(P, R) = \prod_i R(\alpha_i) = (-1)^{\deg P \cdot \deg R} \prod_j P(\beta_j)$. This is the "cross-evaluation" product, which reduces to the Pandrosion diagonal product when $R = P'$.

5 Summary: the Pandrosion invariant

Invariant	Pandrosion formula	Paper
$P'(\alpha_k)$	$Q(\alpha_k, \alpha_k)$	14
$E_P(\alpha_k)$	$Q(\alpha_k, \alpha_k)^2$	20
$D(P)$	$(-1)^{d(d-1)/2} \det \mathbf{Q}$	22
$D(P)^2$	$\prod E_P(\alpha_k)$	22
$\mathrm{Res}(P, P')$	$\det \mathbf{Q}$	22

The discriminant is the determinant of the Pandrosion interaction matrix, and its square is the product of all Pandrosion energies at the roots. This closes the circle between the pointwise theory (Papers 14, 15, 20) and the global invariants of the polynomial.

References

[1] I. Besevic, *Derivative-free Gauss–Lucas via Pandrosion influence weights*, Preprint (2026).

[2] I. Besevic, *Laguerre's inequality from Pandrosion energy*, Preprint (2026).

[3] I. Besevic, *The Pandrosion power tower*, Preprint (2026).